

Red Supergiants in M31: Investigating Binary Candidates in Cluster and Field Environments

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ABSTRACT

Red supergiants (RSGs) are a short-lived evolutionary phase of massive stars and provide important constraints on stellar evolution, mass loss, and binary interactions. Ultraviolet (UV) excess can indicate the presence of a hot companion to an RSG. Stellar clusters offer age- and extinction-constrained environments for studying RSG evolution. We investigate the RSG population of the Andromeda Galaxy (M31) using the Panchromatic Hubble Andromeda Treasury (PHAT) survey, which provides resolved photometry spanning the ultraviolet through near-infrared. We identify RSGs associated with stellar clusters in the Andromeda Project cluster catalog using spatial membership and age-dependent color and magnitude selection, yielding 252 RSGs across 237 clusters. We additionally construct a PHAT-wide RSG catalog using near-infrared and optical color-magnitude criteria, containing 5,188 RSGs, of which 11.6% are binary candidates identified through UV excess. Spatially cross-matching this catalog with the cluster catalog identifies 99 cluster-associated RSGs of which 25.3% show UV excess, consistent with the 25.8% fraction found for the cluster RSG sample. The higher binary-candidate fraction of cluster-associated RSGs relative to the field population suggests that stellar environment may be associated with differences in the binary properties of evolved massive stars.

Keywords: Red supergiant stars (1375) — Binary stars (154) — Andromeda galaxy (39) — Star clusters (1567) — Stellar astronomy (1583)

1. INTRODUCTION

Massive stars play a fundamental role in the evolution of galaxies. Through their intense radiation, stellar winds, and eventual supernova explosions, they inject energy into and chemically enrich the interstellar medium, driving subsequent generations of star formation. As the progenitors of neutron stars and black holes, massive stars are also responsible for some of the most energetic phenomena in the universe, including core-collapse supernovae and compact object mergers that result from binary stellar evolution.

Most massive stars with initial masses of approximately $8\text{--}30 M_{\odot}$ evolve through a red supergiant (RSG) phase following the exhaustion of hydrogen in their cores. During this evolutionary stage, RSGs become cool, highly luminous stars with radii hundreds to more than a thousand times that of the Sun. As one of the final evolutionary stages before core collapse, RSGs provide valuable constraints on massive star evolution, mass loss, and the progenitors of Type II supernovae.

The masses, rotation rates, and evolutionary lifetimes of massive stars can be significantly altered by mass

transfer, common-envelope evolution, and mergers, all of which are consequences of binary interactions. Therefore, determining the binary properties of RSGs is essential for understanding the late stages of massive star evolution. However, the binary fraction of RSGs, particularly in different stellar environments, remains poorly constrained. The shared evolutionary history of massive stars and their companions makes stellar clusters an ideal environment for studying RSGs and their potential companions.

Massive stars frequently occur in binary systems with companions of comparable mass, making massive main-sequence stars natural candidates for RSG companions. Since the evolutionary timescale of a massive star depends strongly on its mass, the initially more massive component evolves first and can enter the short-lived RSG phase while its companion remains on the main sequence. As a result, RSG+RSG systems are expected to be relatively uncommon, since both stars would need to be observed during the brief RSG phase at the same time. Instead, an RSG is more likely to have a slightly less massive B- or OB-type companion that has not yet

evolved off the main sequence. The two stars also have distinct spectral energy distributions (SED): the cool RSG is bright at optical and infrared wavelengths, while the hot B- or OB-type companion emits strongly at ultraviolet wavelengths, allowing the two components to be distinguished observationally. The prevalence and properties of such binary systems may also depend on the environments in which massive stars form.

Star clusters are dense groupings of stars that formed from the same molecular cloud. As a result, the stars within a cluster share similar ages and chemical compositions, making these systems valuable laboratories for studying stellar evolution in environments with well-constrained ages and dust extinction. Their common formation history also allows cluster ages to be estimated through isochrone fitting, providing an age estimate for the stellar population as a whole. These age estimates allow RSGs in clusters to be studied not only as a function of their environment, but also as a function of the age of their host cluster, allowing us to investigate how the properties of RSGs change over the course of their evolution. Star clusters also provide insight into the initial mass function (IMF), star formation, and the evolution of stellar populations. Comparing RSGs residing in stellar clusters with those in the surrounding field therefore provides an opportunity to investigate whether local environment influences the binary properties and evolution of massive stars. However, to fully characterize these processes, individual cluster members must be spatially resolved. Nearby galaxies in the Local Group provide an ideal opportunity to observe star clusters across an entire galaxy while avoiding much of the dust extinction that limits observations within the Milky Way.

In this work, we investigate whether the environments in which massive stars form within the Andromeda galaxy are associated with differences in the evolutionary properties of RSGs. Specifically, we examine whether RSGs located in stellar clusters are more likely to exhibit ultraviolet excess, a potential signature of a hot binary companion, as suggested by [M. Dai et al. \(2025\)](#) and [K. F. Neugent \(2021\)](#). By comparing the binary candidate fractions of cluster and field RSG populations, we investigate whether dense stellar environments are associated with differences in the binary properties of evolved massive stars.

Section 2 summarizes the PHAT survey, the AP cluster catalog, and the methods used to access the data used in this paper. In Section 3, we describe our method for identifying RSGs within stellar clusters using selection criteria in the optical color-magnitude diagram (CMD), allowing us to construct a cluster RSG sam-

ple from the AP cluster catalog. Section 4 describes the construction of the PHAT-wide field RSG catalog, including the selection criteria and color cuts used to identify RSG candidates and establish a field RSG population for comparison with cluster RSGs. Section 5 then describes our criteria for identifying RSGs in binary systems using UV excess. In Section 6, we present the properties of the cluster and field RSG populations and compare their respective binary candidate fractions. Finally, in Section 7, we discuss the implications of our results and compare our RSG and binary candidate populations with previous studies of M31 RSGs, including [P. Massey et al. \(2021a\)](#)³ and [M. Dai et al. \(2025\)](#).

2. DATA

To investigate RSG populations in a nearby Local Group galaxy, we utilize observations from the Panchromatic Hubble Andromeda Treasury (PHAT), which covered approximately one-third of the star-forming disk of the Andromeda galaxy (M31). PHAT obtained imaging in six filters spanning the ultraviolet (UV) through the near-infrared (NIR) with the Hubble Space Telescope (HST): F275W and F336W with WFC3/UVIS, F475W and F814W with ACS/WFC, and F110W and F160W with WFC3/IR. This broad wavelength coverage enables detailed studies of stellar populations across a wide range of evolutionary stages while also providing sensitivity to ultraviolet excesses that may indicate the presence of hot binary companions. We accessed the PHAT V3 data through NOIRLab’s Astro Data Lab; the dataset is also available from the MAST archive at [doi:10.17909/T91S30](https://doi.org/10.17909/T91S30). A complete description of the PHAT survey design is given by [J. J. Dalcanton et al. \(2012\)](#), while the PHAT V3 photometry is described by [B. F. Williams et al. \(2023\)](#).

To identify RSGs associated with stellar clusters, we use the PHAT stellar cluster catalog developed by [L. C. Johnson et al. \(2015\)](#). The catalog was constructed by the Andromeda Project (AP), a citizen science initiative through Zooniverse in which volunteers visually identified stellar clusters in PHAT images. This effort resulted in a catalog of 2,753 stellar clusters that serves as a foundation for studies of M31’s cluster population.

3. CLUSTER RSGS

3.1. Cluster RSG Catalog Selection

To identify RSGs within the stellar clusters of the AP Cluster Catalog, we first identify the PHAT stars spatially associated with each cluster. Using the clus-

³ Analysis of [P. Massey et al. \(2021a\)](#) is reserved for future work.

ter coordinates and aperture radius provided by the AP catalog, we select all PHAT stars that fall within each cluster aperture. This provides a sample of stars associated with each cluster from which we can identify the RSG population.

We then use an optical color-magnitude diagram (CMD) to identify RSGs within each cluster sample. For each cluster, we construct a CMD using the F475W–F814W color and F814W magnitude. This combination of filters allows us to distinguish the cool, luminous RSG population from other stellar populations based on their optical colors and magnitudes. The RSG selection region is defined using age-matched isochrone models (P. Marigo et al. 2008; L. Girardi et al. 2010), and is described below.

Although Section 4.1 discusses the advantages of the NIR CMD for identifying RSGs, we use the optical CMD for our cluster selection for two primary reasons. First, the AP cluster catalog was fit using the Padova isochrone models (P. Marigo et al. 2008; L. Girardi et al. 2010) in the optical CMDs. Defining our selection region using the same filters as those used to fit the cluster isochrones ensures consistency between the cluster properties and our RSG selection. Second, we use the optical CMD to identify the full helium-burning population, including blue supergiants (BSGs). BSGs are comparatively faint in the NIR and would therefore be more difficult to identify using the NIR CMD, potentially biasing the sample against these stars.

Additionally, we restrict the analysis to clusters with $\log(\text{age/yr}) \leq 8.0$ for several reasons. First, this age range encompasses the RSGs of interest in this study, including stars with initial masses of approximately 5–6, $M_{\odot}$ and above, based on the Padova isochrone models (P. Marigo et al. 2008; L. Girardi et al. 2010). Restricting the sample to young clusters therefore allows us to focus on RSGs produced by relatively massive stars while minimizing contamination from older stellar populations. Additionally, RSG identification becomes increasingly difficult at older ages as the RSG region of the optical isochrone begins to overlap with the region of background field red giant branch (RGB) stars. This overlap makes it more difficult to distinguish RSGs from lower-mass evolved stars based on photometry alone. Finally, this age restriction allows us to investigate RSGs at lower luminosities, down to approximately $\log(L/L_{\odot}) = 3.5$, extending below the $\log(L/L_{\odot}) \sim 4$ threshold commonly adopted in previous studies. This provides an opportunity to explore a broader population of RSGs while maintaining a CMD region in which they can be reliably identified.

The observed CMD is compared to Padova solar-metallicity isochrone models (P. Marigo et al. 2008; L. Girardi et al. 2010) matched to the cluster age. The isochrone is reddened according to the cluster extinction (A_v) given by the AP catalog and shifted by the distance modulus of M31, $\mu = 24.47$. We use the age-matched isochrones to define the luminosity and color boundaries of the RSG selection.

As described above, the separation between RSGs and other evolved stellar populations becomes less distinct at older cluster ages. We therefore use age-dependent offsets to balance the completeness of the RSG selection with contamination from field RGB stars. At younger ages, where the RSG region is more clearly separated from the RGB population, the selection boundaries are more inclusive to avoid unnecessarily excluding RSG candidates. At older ages, where the CMD becomes increasingly crowded and the RSG region overlaps more strongly with RGB stars, the boundaries become more restrictive to reduce contamination. These age-dependent offsets allow the selection criteria to account for the changing CMD morphology across the range of cluster ages.

We define the luminosity limit of the RSG selection using the main-sequence turnoff (MSTO) from the Padova isochrone models (P. Marigo et al. 2008; L. Girardi et al. 2010). The MSTO F814W magnitude is obtained from an age-matched set of stellar models and shifted for extinction and distance modulus. To select stars sufficiently brighter than the MSTO to be consistent with the RSG evolutionary phase, while also accounting for photometric errors and potential uncertainties in the cluster ages from the Andromeda Project, we apply an age-dependent offset,

$$\Delta_{\text{MSTO}} = 2.5 - 0.5 [\log(\text{age/yr}) - 8.5], \quad (1)$$

such that the faint boundary of the RSG selection is

$$\text{F814W}_{\text{RSG}} = \text{F814W}_{\text{MSTO}} - \Delta_{\text{MSTO}}. \quad (2)$$

We additionally use the red giant branch tip (RGBt) evolutionary stage from the Padova isochrone models (P. Marigo et al. 2008; L. Girardi et al. 2010) as a reference for the redward limit of the RSG population. The RGBt provides a convenient redward boundary beyond which we do not expect RSGs to be present, allowing us to define the extent of the RSG selection region in the CMD. As the separation between RSGs and other evolved stellar populations becomes less distinct with increasing age, contamination from stars that are too red to be RSGs, including potential asymptotic giant branch (AGB) stars, becomes increasingly important.

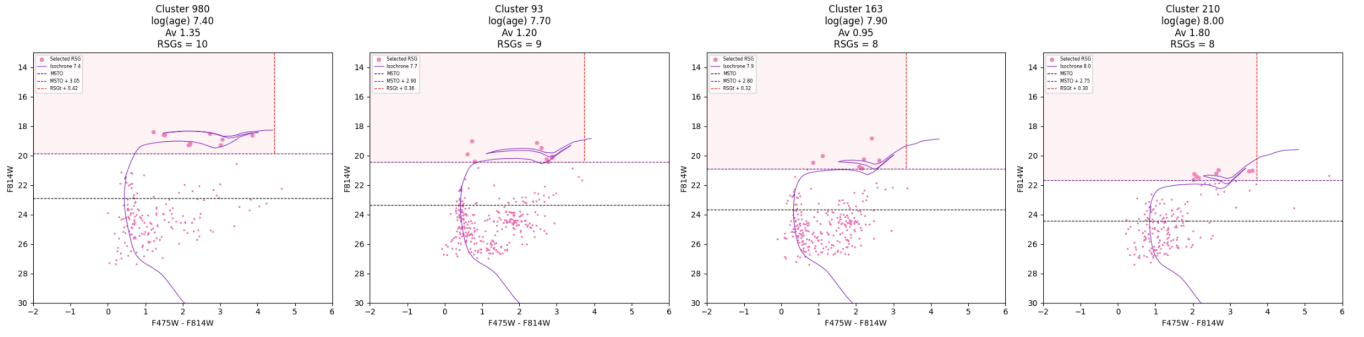


Figure 1. Color-magnitude diagrams for four clusters with large numbers of RSG candidates, ordered by increasing cluster age from left to right. The shaded region shows the selection region used to identify RSG candidates, defined by the MSTO luminosity limit (purple) and the RSG color boundary (red). Selected RSG candidates are shown with larger markers, while the isochrone, MSTO, MSTO offset, and RSG boundary are shown for reference.

We therefore shift this boundary blueward with an age-dependent offset to reduce contamination while retaining as much of the RSG population as possible. The F475W-F814W color of the RGBt point is first reddened according to the cluster extinction, after which an age-dependent color offset is applied:

$$\Delta_{\text{RGBt}} = 0.6 - 0.2 [\log(\text{age/yr}) - 6.5]. \quad (3)$$

The resulting color boundary is therefore

$$\begin{aligned} (F475W - F814W)_{\text{RSG}} = \\ (F475W - F814W)_{\text{RGBt}} - \Delta_{\text{RGBt}}. \end{aligned} \quad (4)$$

To summarize, RSG candidates are selected as stars that are both brighter than the MSTO-based magnitude boundary and bluer than the RSG color boundary, as shown in Figure 1:

$$\begin{aligned} F814W &\leq F814W_{\text{MSTO}} - \Delta_{\text{MSTO}}, \\ F475W - F814W &\leq \\ (F475W - F814W)_{\text{RGBt}} - \Delta_{\text{RGBt}}. \end{aligned} \quad (5)$$

These selection criteria are applied independently to each cluster using its age and A_V , producing a sample of cluster-associated RSG candidates that can be compared with the full PHAT RSG population.

Finally, we require a signal-to-noise ratio (S/N) > 4 in both F475W and F814W to ensure that the photometry used in the RSG selection is sufficiently reliable. These filters define the optical color used in our CMD selection, so low-S/N measurements in either band could introduce large uncertainties in the stellar colors and magnitudes and potentially cause stars to be incorrectly included in or excluded from the RSG selection. We adopt this relatively minimal photometric quality requirement to keep the RSG selection as inclusive as possible while still ensuring that the CMD positions of the selected stars are

reasonably well constrained. Aside from the CMD selection boundaries defined above, the S/N requirements are the only additional photometric quality criteria imposed on the RSG sample.

3.2. Cluster RSG Catalog Results

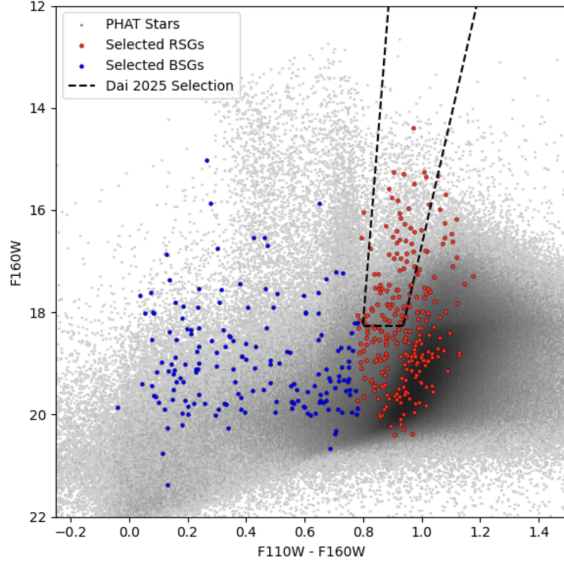
The cluster sample contains 424 evolved massive stars, including 252 RSGs and 170 BSGs⁴, identified across the AP cluster catalog. As shown in Table 1, the RSGs are distributed across 237 clusters, with a maximum of 10 RSGs in a single cluster. While the AP cluster catalog spans $\log(\text{age/yr}) = 6.33\text{--}8.40$ ⁵, the clusters hosting evolved massive stars span $\log(\text{age/yr}) = 6.60\text{--}8.00$, with a mean age of $\log(\text{age/yr}) = 7.71$.

⁴ A total of 424 evolved massive stars were initially identified in the cluster sample. The two stars identified in cluster 1954 lack F110W_{VEGA} detections and therefore cannot be classified as RSGs or BSGs using the infrared color selection discussed later. The final sample consequently contains 422 stars classified as either RSGs or BSGs.

⁵ Of the 1575 clusters considered in the initial sample, 69 were excluded because they lack the necessary F110W and F160W imaging and therefore cannot be analyzed using the infrared RSG selection. These clusters have only optical imaging as a result of the spatially varying filter coverage of the PHAT survey. The excluded cluster IDs are: 46, 74, 155, 197, 255, 261, 303, 346, 358, 371, 387, 394, 400, 500, 505, 611, 654, 668, 748, 764, 771, 777, 1072, 1134, 1135, 1199, 1221, 1247, 1269, 1326, 1462, 1625, 1679, 1768, 1819, 2121, 2312, 2314, 2363, 2384, 2412, 2523, 2565, 2586, 2684, 2894, 3163, 3205, 3320, 3331, 3372, 3452, 3470, 3594, 3637, 3652, 3727, 3735, 3778, 3937, 3955, 3956, 3980, 4083, 4145, 4185, 4259, 4600, 5216. This leaves 1506 clusters in the final sample.

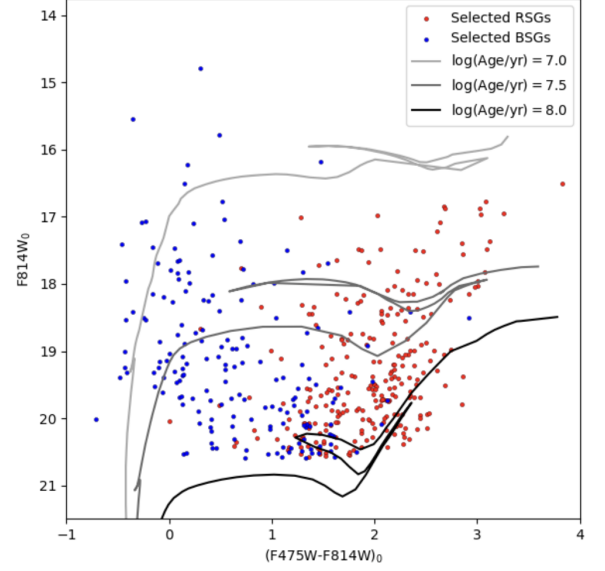
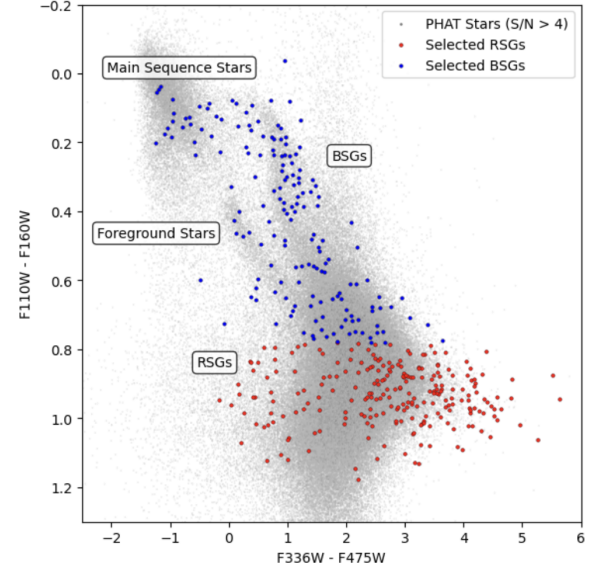
Table 1. Summary of cluster RSG sample.

	HeB Clusters
Number of clusters	237
Total number of HeB stars	424
Number of RSGs	252
Number of BSGs	170
Mean evolved stars per cluster	1.79
Maximum evolved stars per cluster	10
$\log(\text{age/yr})$ range	6.60–8.00
Mean $\log(\text{age/yr})$	7.71

**Figure 2.** Infrared CMD of PHAT stars (gray) and CMD-selected HeB stars, with RSGs shown in red and BSGs in blue. The dashed lines show the RSG selection criteria from Dai et al. (2025), illustrating the difference between our broader selection and the more restrictive criteria used in their RSG catalog.

The BSG and RSG populations occupy distinct regions of the color-magnitude diagram, with RSGs appearing at substantially redder colors than BSGs. We visually inspected the evolved-star populations in each cluster in the $F110W_{\text{VEGA}}-F160W_{\text{VEGA}}$ color versus $F110W_{\text{VEGA}}$ magnitude diagram to identify the separation between the two populations (See Figure 2). This infrared color provides a clear distinction between the cool RSGs and hotter BSGs, as the spectral energy distributions of RSGs peak at longer wavelengths and produce substantially redder NIR colors. Based on this separation, we define the RSG selection boundary as

$$F110W_{\text{VEGA}} - F160W_{\text{VEGA}} \geq 0.78. \quad (6)$$

**Figure 3.** Dereddened optical CMD of cluster selected HeB stars. RSGs are indicated in red and BSGs in blue. Pavova isochrones (L. Girardi et al. (2010) & P. Marigo et al. (2008)) of $\log(\text{age/yr}) = 7.0, 7.5, 8.0$ are over plotted for reference.**Figure 4.** Color-color diagram of PHAT stars (gray) and CMD-selected HeB stars, with RSGs shown in red and BSGs in blue. Main-sequence stars, foreground stars, BSGs, and RSGs are labeled to show the separation between the major stellar populations.

Both the BSG and RSG regions may also contain yellow supergiants (YSGs); however, these stars are difficult to isolate based on color alone because they are relatively rare and occupy this evolutionary phase for a short period of time.

The resulting age distribution shows a clear increase in the number of RSGs toward older clusters, as shown in Table 2. No RSGs are identified in the youngest age bin ($\log(\text{age/yr}) = 6.50\text{--}6.75$), although two BSGs are present. These stars may be mass gainers or blue stragglers, which can remain on the main sequence for longer than the rest of the cluster population while remaining sufficiently luminous to fall within the BSG selection region. Four RSGs are then identified in the $\log(\text{age/yr}) = 6.75\text{--}7.00$ bin, with the RSG population increasing substantially toward older ages.

The number of RSGs then increases to 38, 57, and 146 in the 7.25–7.50, 7.50–7.75, and 7.75–8.00 bins, respectively. This trend is also reflected in the number of RSG-hosting clusters, which rises from 4 in the 6.75–7.00 bin to 84 in the oldest bin. (see bottom right panel of Figure 9) Thus, the majority of the cluster RSG sample is concentrated at $\log(\text{age/yr}) \gtrsim 7.5$, with the oldest age bin alone containing approximately 58% of the identified RSGs. The sample also contains 170 BSGs, providing a comparison population for examining the relative occurrence of these evolved massive-star populations across cluster ages.

4. COMPARISON TO FIELD POPULATIONS

4.1. PHAT-wide RSG Catalog Selection

To identify RSG candidates within the PHAT survey, we use NIR photometry, where RSGs are among the brightest stellar populations due to their high luminosities and cool effective temperatures. The initial selection is based on the color-magnitude criteria developed by M. Dai et al. (2025), which isolate luminous cool stars in the M31 disk while reducing contamination from other evolved stellar populations. RSG candidates are selected using the following color-magnitude criteria:

$$\begin{aligned}
 &\text{left boundary:} \\
 &\quad F160W_{\text{VEGA}} - 18.271 \geq \\
 &\quad -71.250 (F110W_{\text{VEGA}} - F160W_{\text{VEGA}} - 0.799), \\
 &\text{right boundary:} \\
 &\quad F160W_{\text{VEGA}} - 18.271 \leq \\
 &\quad -25.000 (F110W_{\text{VEGA}} - F160W_{\text{VEGA}} - 0.937), \\
 &\text{bottom boundary:} \\
 &\quad F160W_{\text{VEGA}} \leq 18.271.
 \end{aligned} \tag{7}$$

These criteria identify stars occupying the expected RSG region of the $F110W_{\text{VEGA}} - F160W_{\text{VEGA}}$ CMD while limiting the sample to the most luminous cool stars.

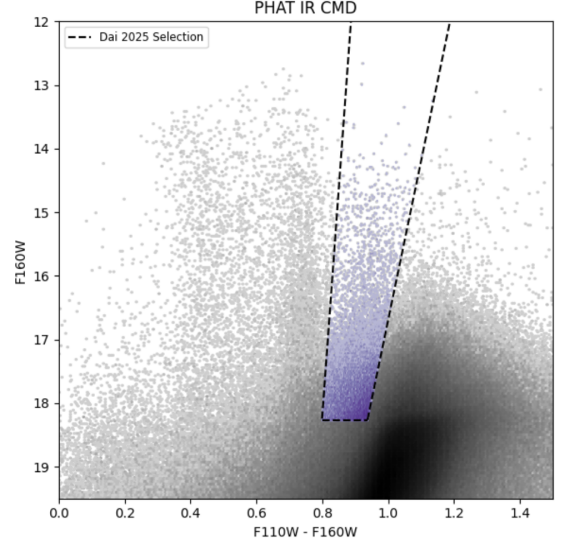


Figure 5. Observed color-magnitude diagram for PHAT. Dashed lines show the M. Dai et al. (2025) boundaries defining the RSG selection region. Purple stars indicate the stars selected within these boundaries.

To reduce contamination from asymptotic giant branch (AGB) stars, which can overlap with RSGs in color-magnitude space, we apply an additional luminosity-dependent cutoff inspired by M. Dai et al. (2025):

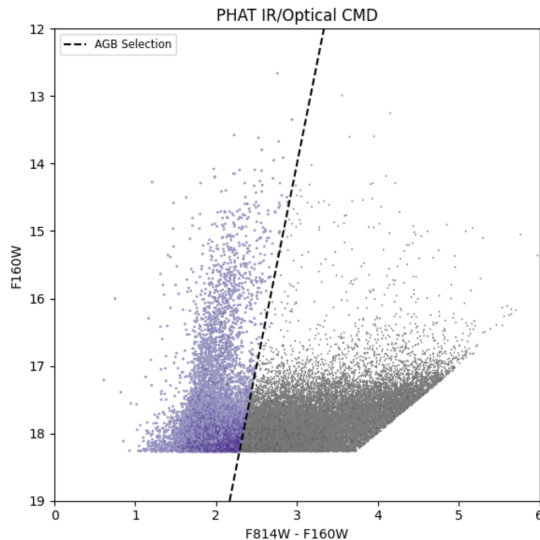
$$\begin{aligned}
 F160W_{\text{VEGA}} &\leq \\
 &-6.00(F814W_{\text{VEGA}} - F160W_{\text{VEGA}}) + 32.00.
 \end{aligned} \tag{8}$$

To ensure reliable photometry, additional quality cuts are applied. Following M. Dai et al. (2025), we require stars to have small photometric uncertainties in the relevant PHAT filters: $F814W_{\text{ERR}}, F110W_{\text{ERR}}, \text{ and } F160W_{\text{ERR}} \leq 0.01$ and $F475W_{\text{ERR}} \leq 0.2$. We also require valid GST photometry flags in the optical and NIR filters: $F475W_{\text{GST_FLAG}}, F814W_{\text{GST_FLAG}}, F110W_{\text{GST_FLAG}}, \text{ and } F160W_{\text{GST_FLAG}} = 1$. However, we do not apply the $F275W_{\text{GST_FLAG}} = 1$ and $F336W_{\text{GST_FLAG}} = 1$ requirements used by M. Dai et al. (2025).

The GST (“good star”) catalogs prioritize sources with high-quality photometry, requiring high signal-to-noise detections and low crowding in both filters of a given camera (J. J. Dalcanton et al. 2012). These strict criteria can introduce incompleteness, particularly in the F275W and F336W filters, which are significantly shallower than the optical and NIR filters for typical RSG colors. As a result, many high-quality RSG candidates lack GST detections in these UV bands. Applying UV

Table 2. Distribution of clusters and RSGs as a function of cluster age.

Age	Mean log(age/yr)	Clusters	Total Mass in Clusters ($M_{\odot}$)	RSGs	RSG-Hosting Clusters
6.50–6.75	6.637	107	1.11×10^5	0	0
6.75–7.00	6.898	102	1.06×10^5	4	4
7.00–7.25	7.153	74	1.03×10^5	7	6
7.25–7.50	7.407	135	2.09×10^5	38	21
7.50–7.75	7.652	142	2.49×10^5	57	34
7.75–8.00	7.914	280	5.30×10^5	146	84

**Figure 6.** Observed color-magnitude diagram for PHAT showing the overlap between red supergiants (RSGs) and asymptotic giant branch (AGB) stars across optical and infrared wavelengths, as inspired by [M. Dai et al. \(2025\)](#). The plotted stars are the RSG candidates selected in Figure 5, with the selection boundary separating the RSG and AGB populations and the purple region indicating the RSG candidates.

GST requirements would preferentially remove sources with faint UV measurements and reduce sample completeness. Instead, we retain GST requirements for the optical and NIR filters, where RSGs have stronger detections and reliable photometry is critical for identifying the RSG population.

The crowding requirement can also bias the sample against RSGs located within stellar clusters. The crowding measurement quantifies the extent to which nearby sources contribute to the measured flux of a star, with higher values indicating that the photometry is more strongly affected by neighboring sources. Because stellar clusters contain high densities of stars, RSGs within clusters are more likely to have elevated crowding measurements and consequently fail the GST criteria. Applying a strict crowding cut would therefore preferen-

tially exclude cluster RSGs and bias comparisons between cluster and field populations.

Finally, we restrict our sample to the PHAT star-forming disk region of M31, where both the RSG and stellar cluster populations are well characterized. This restriction minimizes the effects of highly crowded photometric environments and ensures that the resulting sample is representative of the star-forming disk population.

4.2. PHAT-wide RSG Catalog Results

In order to directly compare the RSG population in clusters to field RSGs, we must first apply the same selection criteria to both populations. The cluster RSG sample and PHAT-wide RSG sample cannot be directly compared as cluster and field populations because they were selected using different criteria (see Figure 2). To make a consistent comparison, we use the PHAT-wide RSG selection criteria for both cluster and field stars. We identify PHAT-wide RSGs that fall within the aperture radii of AP clusters and define these as the PHAT-wide cluster RSG sample. From the PHAT-wide RSG catalog, we identify 99 RSGs residing in 82 clusters. This provides a consistent sample with which to assess the fraction of the PHAT-wide RSG population associated with clusters.

Our PHAT-wide selection identifies 5,188 RSGs across the PHAT survey, while [M. Dai et al. \(2025\)](#) identified 2,612 RSGs. Applying the same cluster-matching procedure described above to the RSG catalog from [M. Dai et al. \(2025\)](#), we find 81 RSGs residing in 69 clusters. Thus, while both studies identify RSG populations across the PHAT survey and use similar selection approaches, our PHAT-wide sample contains substantially more RSGs and consequently more cluster-associated RSGs. This difference is primarily due to differences in the selection criteria adopted by the two studies, including differences in the treatment of GST flags in the UV filters. In particular, our selection allows GST-flagged sources in the UV filters, whereas [M. Dai et al. \(2025\)](#) does not. Therefore, the PHAT-wide sample is not intended to reproduce the [M. Dai et al. \(2025\)](#) catalog, but instead provides an independently selected RSG popu-

lation using criteria inspired by their work. These differences in selection criteria are important when comparing the numbers and cluster fractions of the two samples.

This leaves us with three separate RSG samples: the cluster RSG sample, selected using the age-dependent CMD criteria developed for each cluster; the PHAT-wide RSG sample, consisting of stars that meet the RSG selection criteria described above and inspired by [M. Dai et al. \(2025\)](#); and the PHAT-wide cluster RSG sample, which is a subset of the PHAT-wide RSG sample containing stars that fall within AP cluster boundaries. The PHAT-wide cluster and field populations can therefore be compared directly using identical selection criteria, while the cluster CMD sample provides an independent measurement of the cluster RSG population. A summary of these results is given in Table 4.

5. BINARY CANDIDATE SELECTION

To identify potential binary RSG systems, we search for ultraviolet (UV) excesses that may indicate the presence of a hot companion ([K. F. Neugent 2021](#); [M. Dai et al. 2025](#)). Red supergiants have cool effective temperatures ($\sim 3500 - 4500$ K) and therefore emit relatively little flux at UV wavelengths. In an unresolved binary system containing an RSG and a hot main-sequence companion, the combined SED can differ significantly from that of an isolated RSG. Although the RSG dominates the optical and infrared emission, the hot companion contributes a substantial fraction of the UV flux. As a result, the combined system can appear brighter in the F336W and F275W filters than expected for a single RSG, producing a UV excess.

To identify these systems, we examine the UV-optical color-color space developed by [M. Dai et al. \(2025\)](#), where the expected colors of single RSGs occupy a well-defined region. The presence of a hot companion shifts the combined RSG and companion SED toward bluer UV-optical colors, allowing potential binary systems to be separated from the single-star population. Candidate binary systems are selected using the criterion:

$$\begin{aligned} F336W_{\text{VEGA}} - F814W_{\text{VEGA}} \leq \\ 3.00(F814W_{\text{VEGA}} - F160W_{\text{VEGA}}) - 0.25. \end{aligned} \quad (9)$$

Additionally, we require a signal-to-noise ratio of $F336W_{\text{SNR}} > 4$ to ensure that the UV detection is statistically significant. This threshold reduces the likelihood that photometric uncertainties or noise fluctuations are misidentified as UV excesses, which is particularly important because RSGs are intrinsically faint at UV wavelengths. Due to the large number of non-detections in the F275W filter, we restrict the binary-candidate selection to F336W UV excesses. This provides a more

complete and consistent sample while avoiding biases associated with requiring F275W detections from the shallower UV observations.

Only RSG candidates satisfying the full photometric quality criteria described in Section 4.1 for the PHAT-wide sample and Section 3.1 for the cluster RSG sample are considered for UV-excess selection.

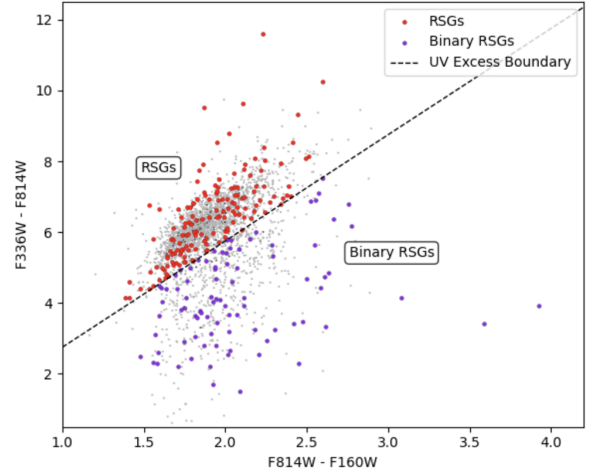


Figure 7. Color-color diagram of CMD-selected RSGs showing the UV-excess binary candidates. Single RSGs are shown in red and binary candidates in purple. The PHAT-wide RSG sample is shown in grey for comparison. The dashed line marks the boundary used to identify UV excess and is inspired by the UV excess plots of [Dai et al. \(2025\)](#).

Table 3. Binary-candidate fractions for the PHAT RSG samples.

Sample	Binary Fraction (%)
Cluster RSGs	25.8
PHAT-wide RSGs: Clusters	25.3
PHAT-wide RSGs: Field	11.6

6. RESULTS

Table 4 summarizes the three RSG populations created in this paper. The cluster RSG sample contains 252 RSGs across 149 clusters, while the PHAT-wide RSG sample contains 5,188 RSGs across the full PHAT survey, of which 99 are located within 82 AP cluster apertures. For comparison, the [M. Dai et al. \(2025\)](#) catalog contains 2,612 RSGs across the PHAT survey, with 81 RSGs residing in 69 clusters when matched to the AP cluster catalog using the same procedure. Thus,

Table 4. Summary of RSG samples.

	Number of RSGs	Number of Clusters
Cluster RSGs	252	149
PHAT-wide RSGs: Clusters	99	82
PHAT-wide RSGs: All	5188	...
Dai+25 RSGs: Clusters	81	69
Dai+25 RSGs: All	2612	...

the number of cluster-associated RSGs is broadly similar between the two PHAT-wide selections despite the substantially different sizes of their total RSG populations.

Of the 5,188 PHAT-wide RSGs, 99, or approximately 1.91%, fall within AP cluster apertures. The corresponding fraction for the [M. Dai et al. \(2025\)](#) sample is 81 out of 2,612 RSGs, or approximately 3.10%.⁶ The cluster RSG sample contains substantially more RSGs per cluster, with 252 RSGs identified across 149 clusters. These results demonstrate that only a small fraction of the PHAT-wide RSG population is associated with the identified AP clusters.

The cluster population also shows a notably different binary-candidate fraction from the field population. Binary candidates are identified using the UV-excess criteria described in Section 5. The binary-candidate fraction among PHAT-wide cluster RSGs is 25.3%, compared to 11.6% for PHAT-wide field RSGs, making the cluster fraction approximately 2.18 times higher.

The binary-candidate fraction also varies with age (Figure 8). For the [M. Dai et al. \(2025\)](#) sample, the binary fraction increases from 11.1% at $\log(\text{age/yr}) = 6.50\text{--}7.25$ to 35.7% at $7.25\text{--}7.50$ and 37.0% at $7.50\text{--}7.75$, before decreasing to 18.2% and 9.1% in the two oldest age bins. The cluster RSG sample follows a similar increase at young ages, with binary fractions of 27.3% and 40.7% in the first two age bins, respectively. The binary fraction then decreases to 30.9% at $\log(\text{age/yr}) = 7.50\text{--}7.75$ and 24.1% at $7.75\text{--}8.00$. The decrease toward older ages in both catalogs is notable and may be caused by age-dependent completeness in the binary selection, where the UV-excess criteria may be more sensitive to binary companions in younger populations. Further analysis of the completeness of the binary selection as a function of age is needed to determine whether this trend is physical or the result of observational selection effects.

⁶ The higher fraction of RSGs associated with clusters in the [M. Dai et al. \(2025\)](#) sample (3.10%) compared to the PHAT-wide sample (1.91%) is not expected given the selection criteria and will require further investigation.

The fraction of clusters containing RSGs also varies with age (Figure 9). In the [M. Dai et al. \(2025\)](#) sample, this fraction increases from 6.0% at $\log(\text{age/yr}) = 6.50\text{--}7.25$ to a maximum of 12.0% at $7.50\text{--}7.75$, before declining to 4.6% at $7.75\text{--}8.00$ and 1.1% at $8.00\text{--}8.50$. In contrast, the cluster RSG sample shows an increasing fraction of RSG-hosting clusters across the four age bins, rising from 3.5% in the youngest bin to 15.6%, 23.9%, and 30.0% in the subsequent bins. The number of RSGs in the cluster sample similarly increases across these bins, from 11 to 38, 57, and 146 RSGs, respectively. The increase in the number of RSGs and RSG-hosting clusters with age in the cluster sample is broadly consistent with the expected evolution of stars into the RSG phase. In contrast, the decrease in the [M. Dai et al. \(2025\)](#) sample is less expected and may be related to the bottom selection boundary in Equation 7, which preferentially excludes older RSG populations based on NIR luminosity. Further investigation of the selection boundaries and their effects on the age distribution is needed to determine the origin of this difference.

7. DISCUSSION

The substantially different sizes of the PHAT-wide and [M. Dai et al. \(2025\)](#) RSG catalogs demonstrate the sensitivity of RSG population measurements to the adopted selection criteria. Although our PHAT-wide selection is inspired by the criteria of [M. Dai et al. \(2025\)](#), it is not intended to reproduce their catalog. Differences in screening criteria, including the treatment of GST flags in the UV filters, result in a larger RSG population in our sample. Importantly, applying the same cluster-matching procedure to both catalogs produces broadly similar numbers of cluster-associated RSGs: 99 in 82 clusters in our PHAT-wide sample compared to 81 in 69 clusters in [M. Dai et al. \(2025\)](#). This provides a useful consistency check while also demonstrating that the inferred fraction of RSGs associated with clusters depends on the adopted selection criteria.

The enhanced binary-candidate fraction among cluster RSGs is particularly interesting. The 25.3% fraction in clusters compared to 11.6% in the field suggests that RSGs residing in clusters may have a higher incidence of hot main-sequence companions. Since UV

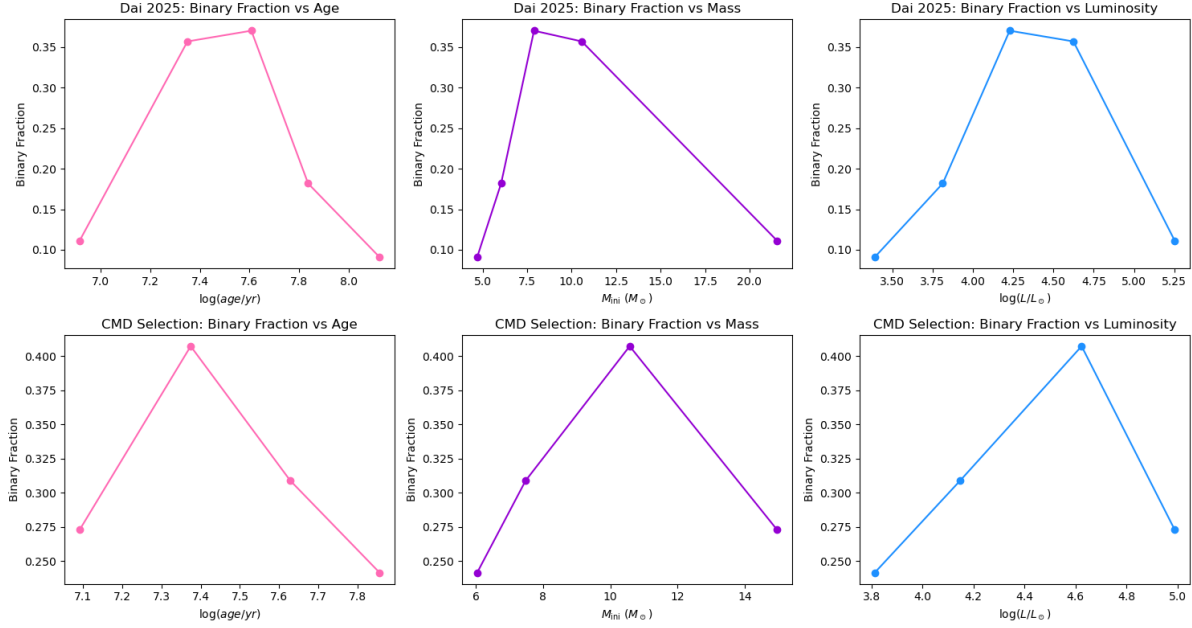


Figure 8. Binary fraction population properties as a function of cluster age for the M. Dai et al. (2025) sample (top) and the CMD-selected cluster RSG sample (bottom). The left panel shows the binary fraction as a function of age (pink), the middle panel shows the binary fraction as a function of mass (purple), and the right panel shows the binary fraction as a function of luminosity (blue). Points show the mean cluster age within each age bin.

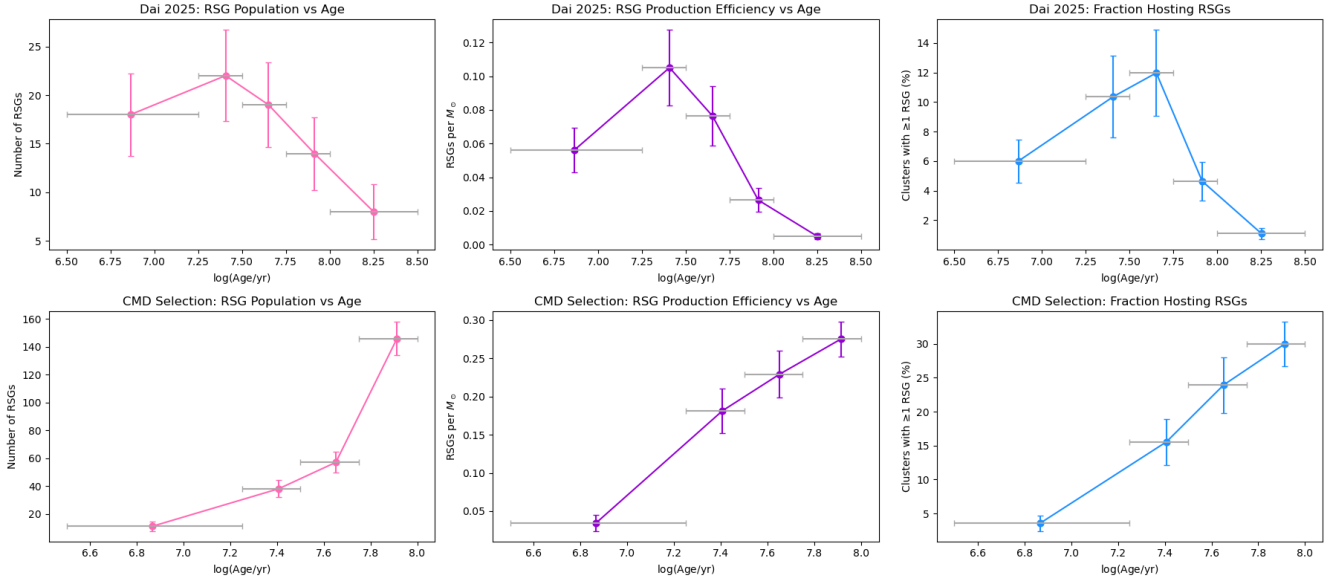


Figure 9. RSG population properties as a function of cluster age for the M. Dai et al. (2025) sample (top) and the CMD-selected cluster RSG sample (bottom). The left panel shows the number of RSGs for each sample (pink), the middle panel shows the RSG production efficiency per unit cluster mass (purple), and the right panel shows the fraction of RSG-hosting clusters with age (blue). Points show the mean cluster age within each age bin with Poisson uncertainties, and grey error bars show bin width.

excess is interpreted as a potential signature of a hot binary companion (K. F. Neugent (2021) & M. Dai et al. (2025)), this difference may indicate that binary evolution plays an important role in the evolution of cluster RSGs. Our UV-excess selection is sensitive to main-

sequence companions with masses of approximately 3–5, $M_{\odot}$ or greater, and therefore does not capture all possible RSG binary companions. However, because the cluster RSG sample and PHAT-wide field sample are selected using different criteria, the difference could also

be influenced by selection effects or differences in the underlying age and mass distributions.

The age-dependent trends provide additional context for this enhanced binary fraction. Both the cluster and [M. Dai et al. \(2025\)](#) samples show their highest binary fractions around $\log(\text{age}/\text{yr}) = 7.25\text{--}7.75$. However, the cluster RSG sample maintains a relatively high binary fraction into the oldest age bin, while the [M. Dai et al. \(2025\)](#) fraction decreases substantially at older ages. This may suggest that the enhanced binary-candidate fraction in clusters is not restricted to a narrow evolutionary stage. Alternatively, the difference may arise from the distinct RSG selection methods and the smaller number of objects in some age bins.

The increasing fraction of RSG-hosting clusters in our cluster sample also suggests a strong age dependence in the occurrence of RSGs. The fraction rises from 3.5% in the youngest bin to 30.0% at $\log(\text{age}/\text{yr}) = 7.75\text{--}8.00$, accompanied by a substantial increase in the number of identified RSGs. This differs from the [M. Dai et al. \(2025\)](#) sample, where the fraction of RSG-hosting clusters peaks at 12.0% around $\log(\text{age}/\text{yr}) = 7.50\text{--}7.75$ before declining at older ages. The increasing number of RSGs in the cluster sample may reflect the evolutionary timescales over which stars of different initial masses enter the RSG phase, while the differences between the two samples may also reflect their different selection criteria.

Overall, these results show that RSG populations in clusters differ from the broader field population both in their binary-candidate fraction and in their age distribution. The higher binary-candidate fraction among cluster RSGs is consistent with an enhanced role for binary evolution, while the strong increase in RSG-hosting clusters toward older ages indicates that the occurrence of RSGs in clusters is strongly age dependent. Separating evolutionary effects from selection and cluster-demographic effects will require further analysis of the age and mass distributions of the cluster and field populations.

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Software: Astropy ([Astropy Collaboration et al. 2013, 2018, 2022](#)), Matplotlib ([J. D. Hunter 2007](#))

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